

AI 4 Earth QMg Presentation

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Introduction

QM9 Dataset

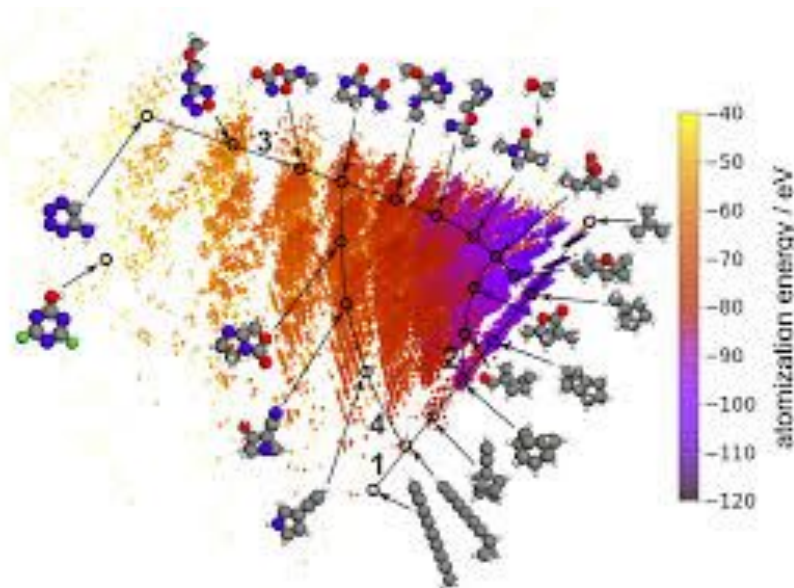
- 134,000 stable organic molecules with up to nine heavy atoms
- 19 regression targets
- Provides geometric, energetic, electronic, and thermodynamic properties

#graphs	#nodes	#edges	#features	#tasks
130,831	~18.0	~37.3	11	19

https://pytorch-geometric.readthedocs.io/en/latest/generated/torch_geometric.datasets.QM9.html

QM9 Dataset

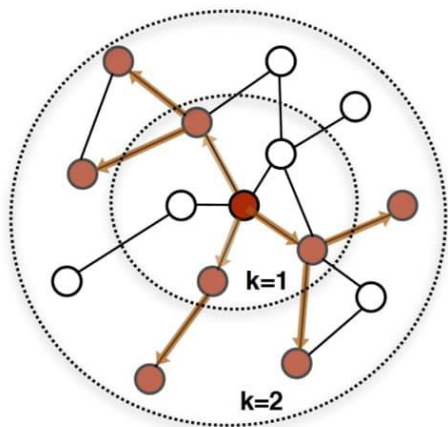
Property	Unit	Description
μ	D	Dipole moment
α	a_0^3	Isotropic polarizability
ϵ_{HOMO}	Ha	Energy of HOMO
ϵ_{LUMO}	Ha	Energy of LUMO
ϵ_{gap}	Ha	Gap($\epsilon_{\text{LUMO}} - \epsilon_{\text{HOMO}}$)
$\langle R^2 \rangle$	a_0^2	Electronic spatial extent
zpve	Ha	Zero point vibrational energy
U_0	Ha	Internal energy at 0 K
U	Ha	Internal energy at 298.15 K
H	Ha	Enthalpy at 298.15 K
G	Ha	Free energy at 298.15 K
C_v	$\frac{\text{cal}}{\text{molK}}$	Heat capacity at 298.15 K



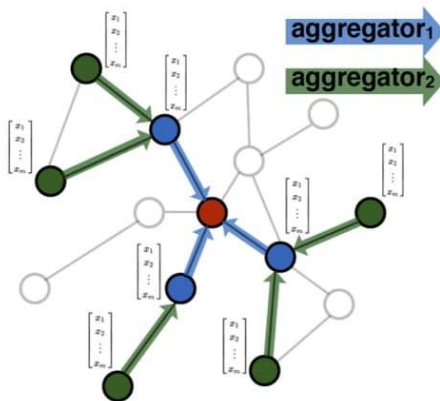
Science-Driven Atomistic Machine Learning - Scientific Figure on ResearchGate. Available from:

https://www.researchgate.net/figure/Visualization-of-the-QM9-dataset-of-small-organic-molecules-using-the-kernel-Principal_fig1_369141971 [accessed 1 Aug 2025]

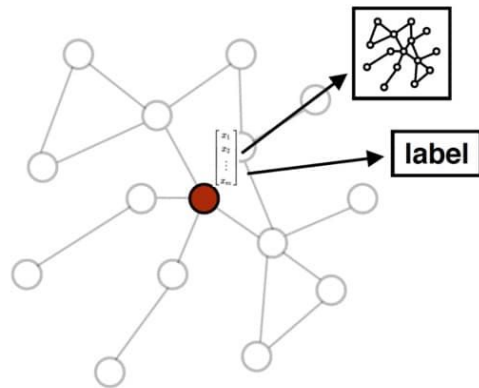
Graph Neural Networks



1. Sample neighborhood



2. Aggregate feature information from neighbors



3. Predict graph context and label using aggregated information

<https://blogs.nvidia.com/blog/what-are-graph-neural-networks/>

Goals

- Gain an understanding of graph neural networks by applying them hands-on
- Replicate the foundational paper: “Neural Message Passing for Quantum Chemistry” by Gilmer et al.
- Try to use state of the art models to beat Neural Message Passing paper using techniques like attention

Methods

- NNConvNet Replication
- Gated Graph Neural Network Replication
- New models with attention
 - Graph attention (node features) - GATconv
 - Graph attention (node features & edge features) - GAT
 - Tower architecture with GATconv2

BAML	BOB	CM	ECFP4	HDAD
GC	GG-NN	DTNN	enn-s2s	enn-s2s-ens5

<https://arxiv.org/pdf/1704.01212>

Results

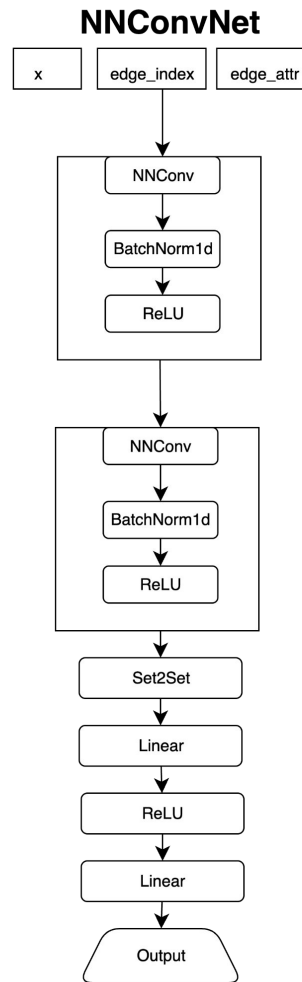
NNConvNet

Main Features:

- Used torch_geometric NNConv
- Added Set2Set global pooling to preserve global structure

Primary Challenge: Overfitting

- Added weight decay (L2 Regularization)
- Used optimizer scheduler
- Added early stopping



NNConvNet | target: dipole moment (μ)

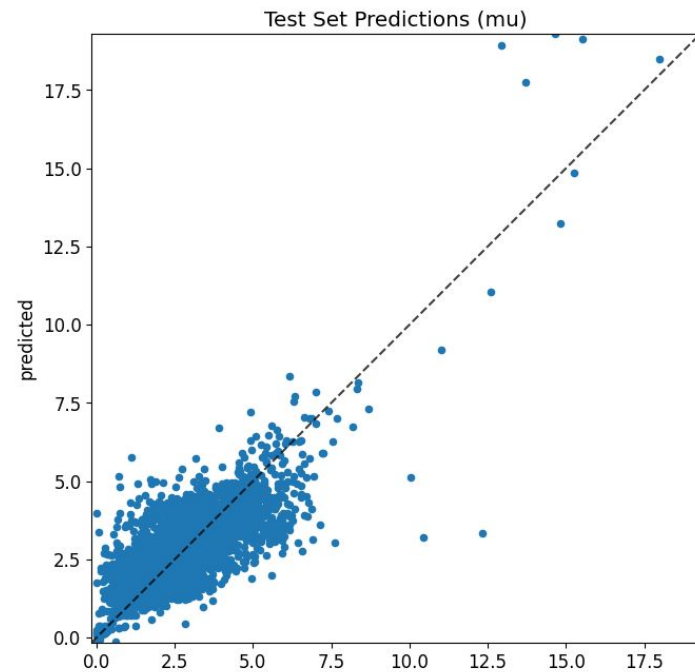
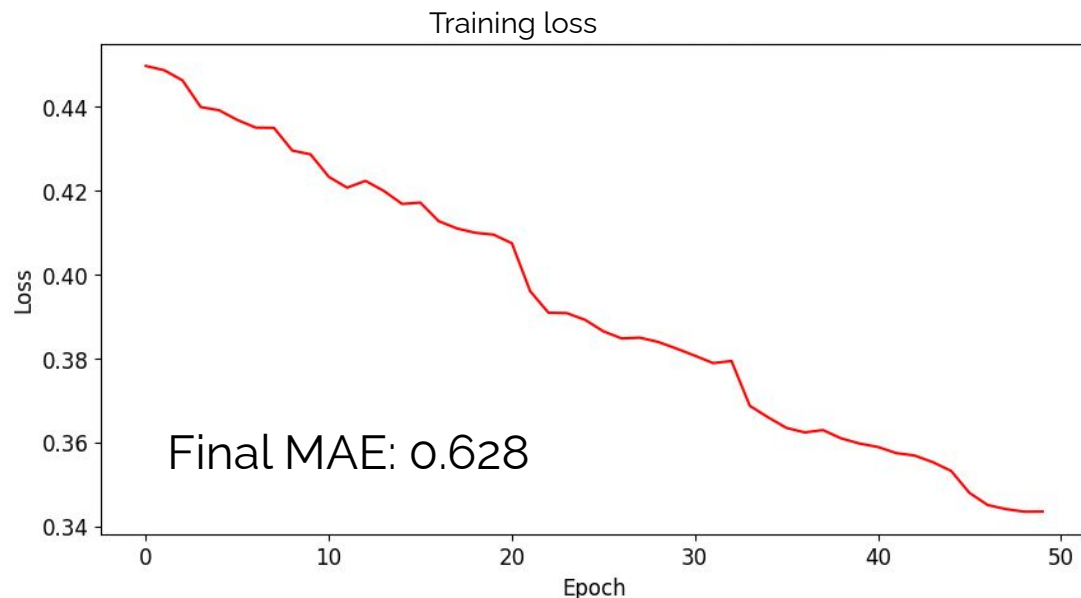
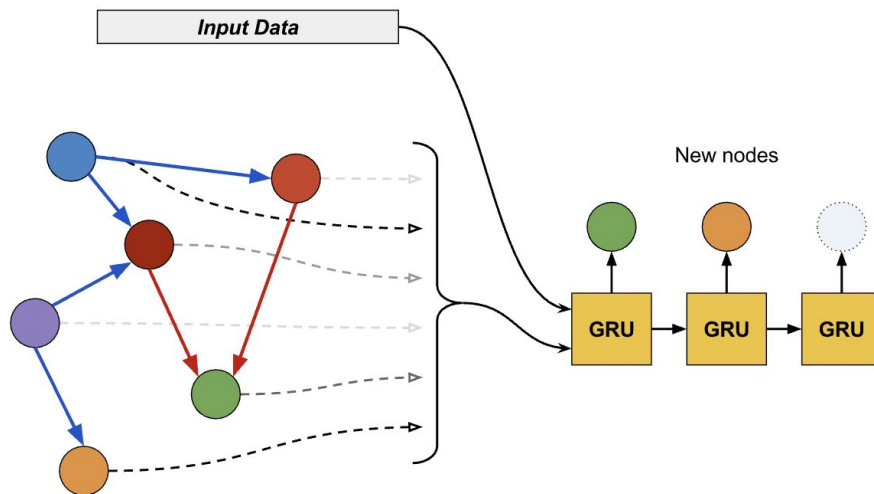


Table 2. Comparison of Previous Approaches (left) with MPNN baselines (middle) and our methods (right)

Target	BAML	BOB	CM	ECFP4	HDAD	GC	GG-NN	DTNN	enn-s2s	enn-s2s-ens5
μ	4.34	4.23	4.49	4.82	3.34	0.70	1.22	-	0.30	0.20

Gated Graph Neural Network

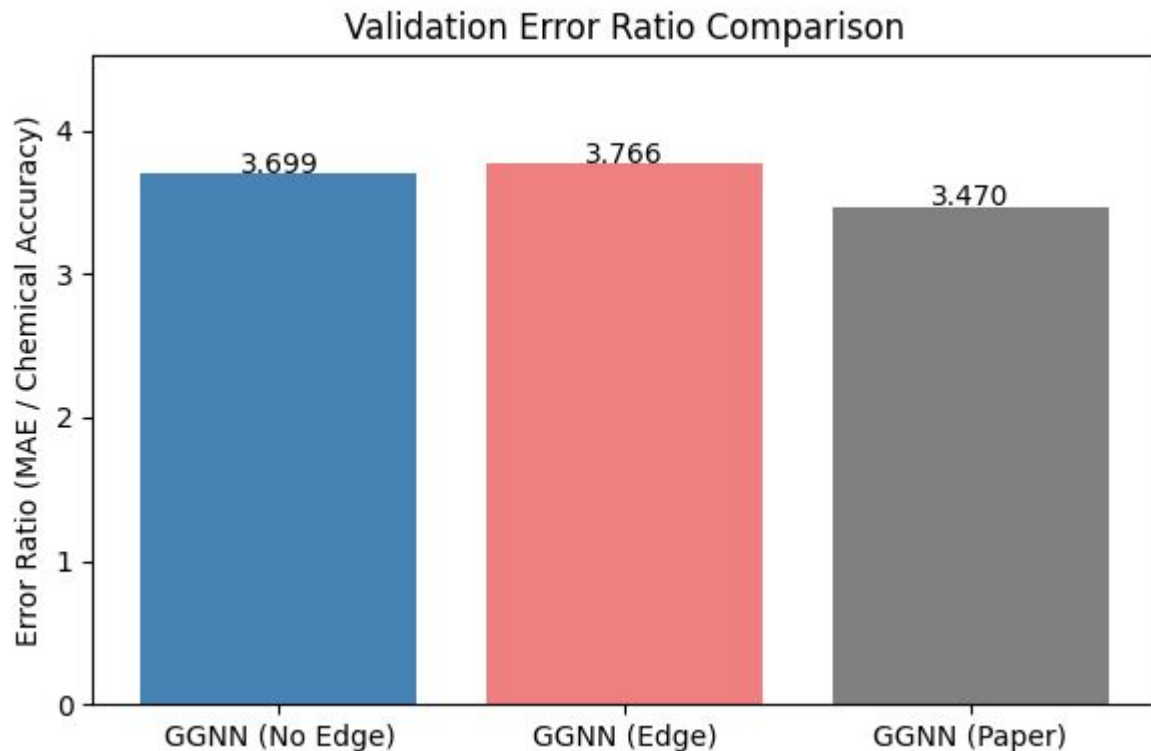


Unroll message passing for a fixed number of step rather than waiting for convergence

Use GRU to update node state

Use Backpropagation Through Time to compute gradients

GGNN: Model Comparison



Target: Dipole Moment (μ).

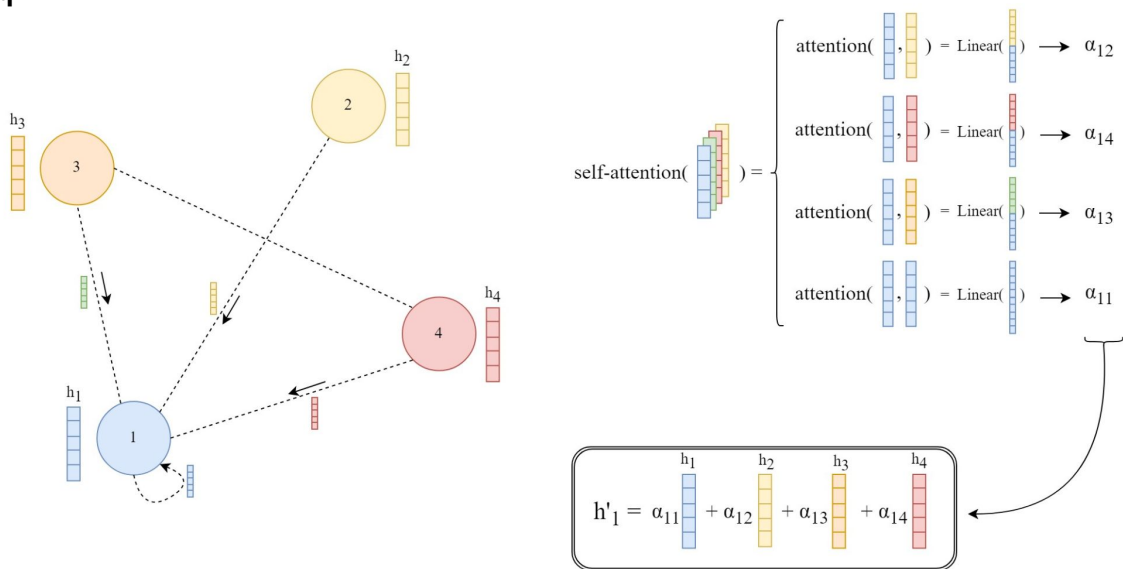
Little different between the model when using edge feature and without edge feature.

Result from our model is relatively close to the result from the paper

Not achieve chemical accuracy (error ratio < 1)

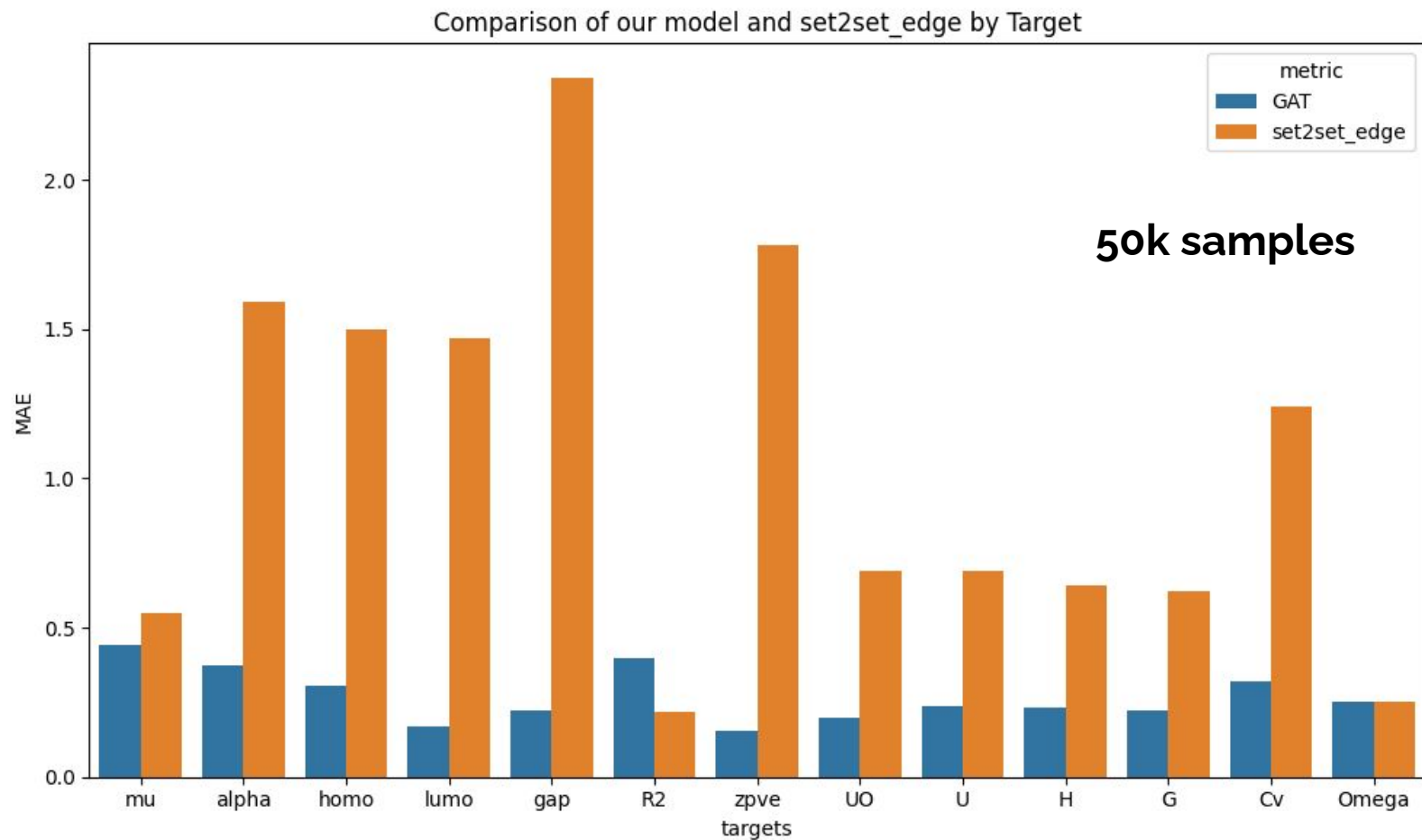
Graph Attention

Graph Attention Networks



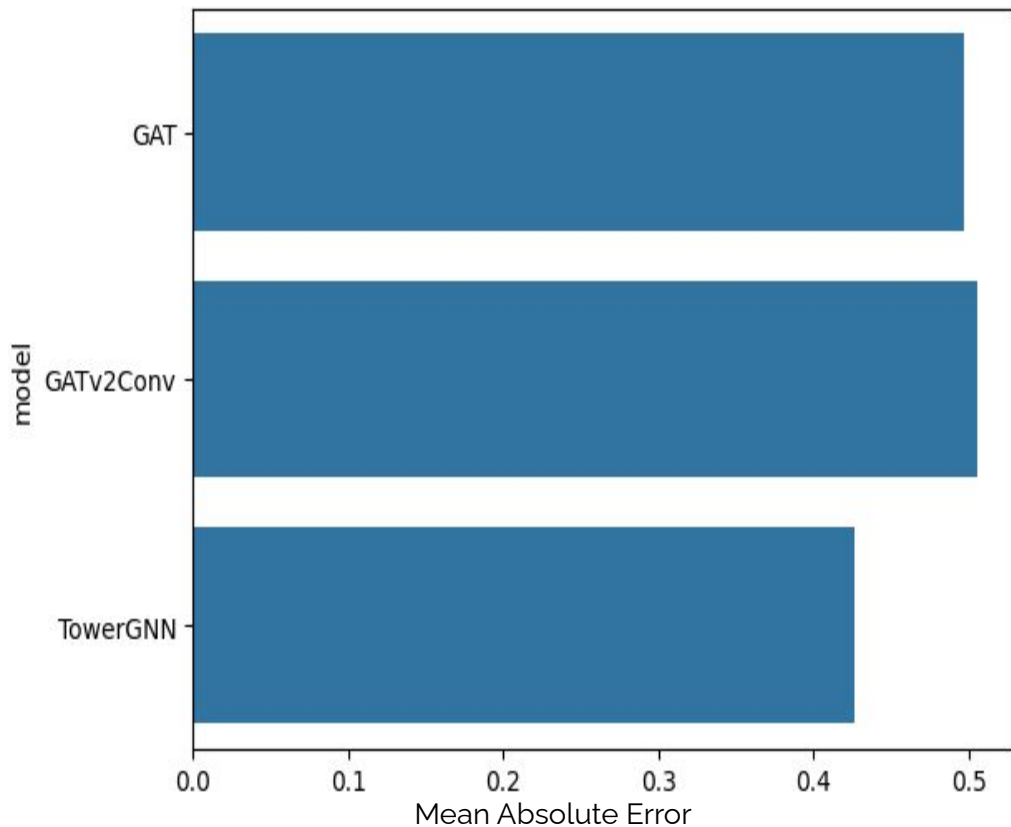
<https://epichka.com/blog/2023/gat-paper-explained//>

Graph Attention vs. Paper's Best Performing Model



Node vs. Feature Reliance

Target: dipole moment (μ) – node feature reliant

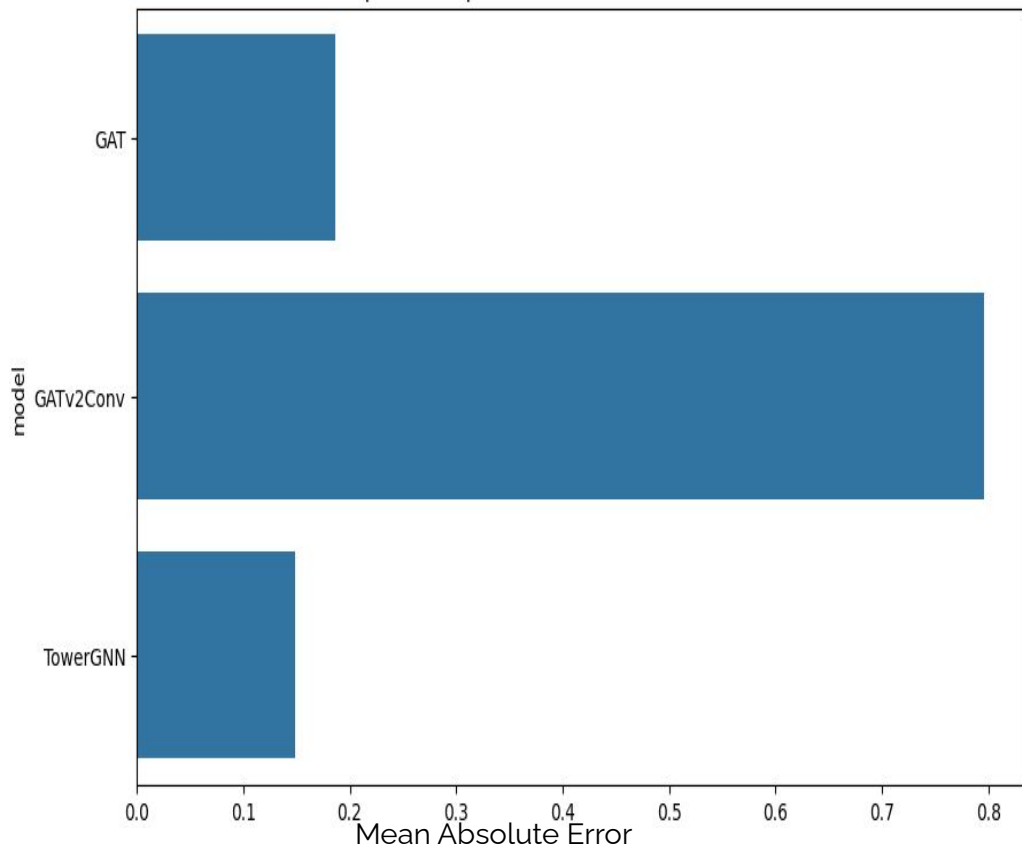


- GAT seems to perform slightly better than GATv2 even though GATv2 is using edge features
- **Hypothesis:**
 - The inclusion of edge feature for training targets that relies heavily on node features might affect performance
 - Same situation for targets that relies heavily on edge features being trained with node features

Node vs. Feature Reliance

Target: Zero Point Vibrational Energy (ZPVE) – edge feature reliant

comparison of performance between models for ZPVE



- Our hypothesis appears to not be true
- But we can't jump to a conclusion given that the Tower architecture with GATv2 outperforms GAT

In **summary**, the bad performance of GATv2 might be due to an architecture mismatch.

Feature Ablation

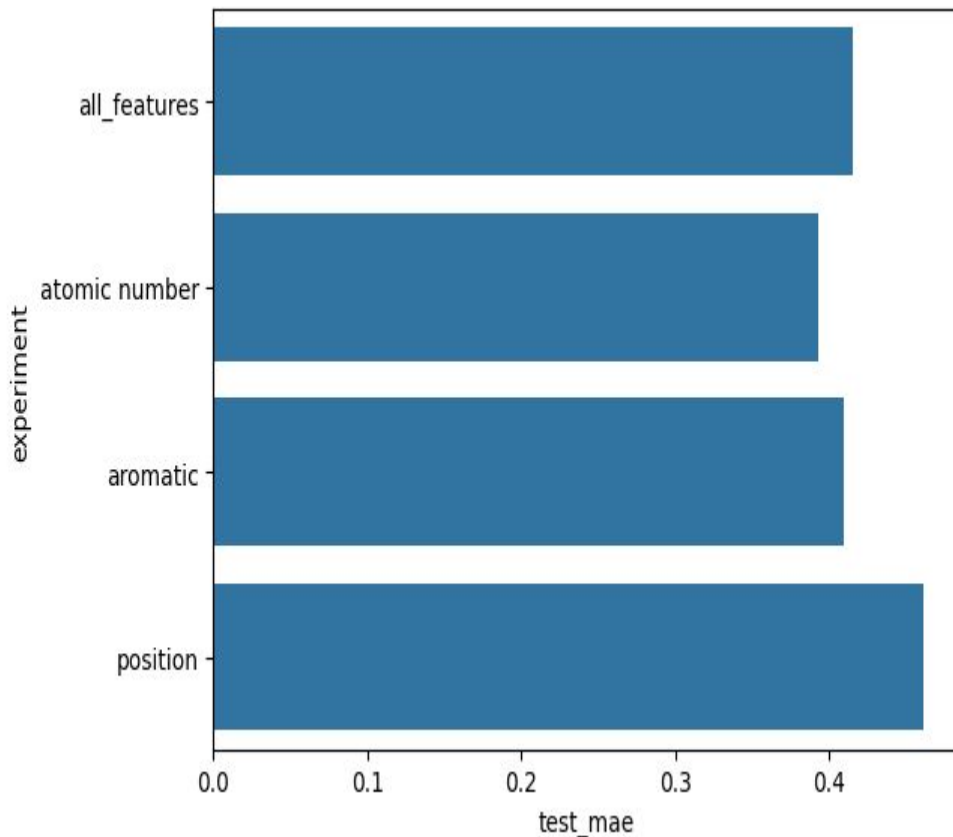


Table 3. Models Trained Without Spatial Information

Model	Average Error Ratio
GG-NN	3.47
GG-NN + Virtual Edge	2.90
GG-NN + Master Node	2.62
GG-NN + set2set	2.57

Table 4. Towers vs Vanilla GG-NN (no explicit hydrogen)

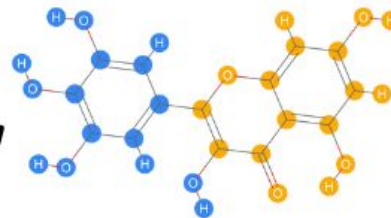
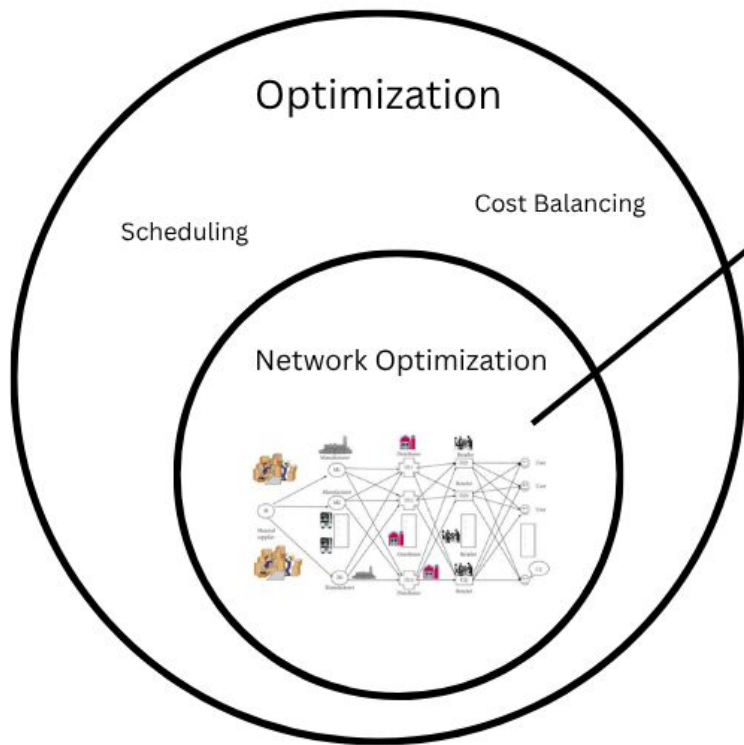
Model	Average Error Ratio
GG-NN + joint training	1.92
towers8 + joint training	1.75
GG-NN + individual training	1.53
towers8 + individual training	1.37

Conclusion: Even with different models, the features importance appears to be relatively the same as discussed in the original paper.

Conclusion

What we learned

Next Steps: Reframing as Network Optimization



Can we use network optimization techniques to infer properties of molecules from the QM9 dataset?

Find optimal dipole moment (μ) given atoms and bonds

Mathematical Formulation of Optimization

$x \in \mathbb{R}^E$: flow over each bond (edges)

A : incident matrix of the graph (from atom-to-bond mapping)

$b \in \mathbb{R}^N$: net charge at each atom

$c \in \mathbb{R}^E$: edge costs (e.g., bond length, bond type)

Optimization Function:

$$\min_x c^\top x \text{ s.t. } Ax = b$$

Thank you!

Any questions?



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